

DSC 291: Learning Theory Assignment 7

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1 Part A: Kernel Methods

1.1 Problem 1. Norm-Based Kernel Bound

For a fixed sample $S = (x_1, \dots, x_n)$,

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_{K,B}) = \mathbb{E}_\sigma \sup_{\|w\| \leq B} \frac{1}{n} \sum_{i=1}^n \sigma_i \langle w, \phi(x_i) \rangle.$$

Pull the inner product inside:

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_{K,B}) = \frac{1}{n} \mathbb{E}_\sigma \sup_{\|w\| \leq B} \left\langle w, \sum_{i=1}^n \sigma_i \phi(x_i) \right\rangle.$$

By Cauchy–Schwarz, the supremum over $\|w\| \leq B$ equals

$$B \left\| \sum_{i=1}^n \sigma_i \phi(x_i) \right\|,$$

so

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_{K,B}) = \frac{B}{n} \mathbb{E}_\sigma \left\| \sum_{i=1}^n \sigma_i \phi(x_i) \right\|.$$

Apply Jensen to $\sqrt{\cdot}$:

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_{K,B}) \leq \frac{B}{n} \sqrt{\mathbb{E}_\sigma \left\| \sum_{i=1}^n \sigma_i \phi(x_i) \right\|^2}.$$

Expand the squared norm:

$$\mathbb{E}_\sigma \left\| \sum_{i=1}^n \sigma_i \phi(x_i) \right\|^2 = \mathbb{E}_\sigma \sum_{i,j=1}^n \sigma_i \sigma_j \langle \phi(x_i), \phi(x_j) \rangle.$$

Since the Rademacher signs are independent and centered,

$$\mathbb{E}[\sigma_i \sigma_j] = 0 \quad \text{for } i \neq j, \quad \mathbb{E}[\sigma_i^2] = 1,$$

so all cross terms vanish:

$$\mathbb{E}_\sigma \left\| \sum_{i=1}^n \sigma_i \phi(x_i) \right\|^2 = \sum_{i=1}^n \|\phi(x_i)\|^2 = \sum_{i=1}^n K(x_i, x_i).$$

Therefore

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_{K,B}) \leq \frac{B}{n} \sqrt{\sum_{i=1}^n K(x_i, x_i)}.$$

Now average over $S \sim \mathcal{D}^n$:

$$\mathfrak{R}_{\mathcal{D},n}(\mathcal{H}_{K,B}) \leq \frac{B}{n} \mathbb{E}_S \sqrt{\sum_{i=1}^n K(x_i, x_i)}.$$

By Jensen again,

$$\mathfrak{R}_{\mathcal{D},n}(\mathcal{H}_{K,B}) \leq \frac{B}{n} \sqrt{\mathbb{E}_S \sum_{i=1}^n K(x_i, x_i)} = \frac{B}{n} \sqrt{n \mathbb{E}[K(X, X)]}.$$

Hence

$$\mathfrak{R}_{\mathcal{D},n}(\mathcal{H}_{K,B}) \leq \frac{B \sqrt{\mathbb{E}[K(X, X)]}}{\sqrt{n}}.$$

1.2 Problem 2. Representer Theorem

Let

$$V := \text{span}\{\phi(x_1), \dots, \phi(x_n)\} \subseteq \mathcal{U}.$$

Take any $w \in \mathcal{U}$ and decompose it orthogonally as

$$w = w_{\parallel} + w_{\perp}, \quad w_{\parallel} \in V, \quad w_{\perp} \perp V.$$

Since every $\phi(x_i) \in V$,

$$\langle w, \phi(x_i) \rangle = \langle w_{\parallel}, \phi(x_i) \rangle + \langle w_{\perp}, \phi(x_i) \rangle = \langle w_{\parallel}, \phi(x_i) \rangle.$$

So the data-fit term in

$$J(w) = \frac{1}{n} \sum_{i=1}^n \text{loss}(\langle w, \phi(x_i) \rangle; y_i) + \lambda \|w\|^2$$

depends only on $w_{\parallel}$.

For the regularizer,

$$\|w\|^2 = \|w_{\parallel}\|^2 + \|w_{\perp}\|^2 \geq \|w_{\parallel}\|^2.$$

Hence

$$J(w) \geq J(w_{\parallel}).$$

Therefore any minimizer can be replaced by one lying in V . So whenever J attains a minimum, some minimizer has the form

$$w^{\star} = \sum_{j=1}^n \alpha_j \phi(x_j)$$

for coefficients $\alpha \in \mathbb{R}^n$.

Now compute the prediction on x_i :

$$\langle w^{\star}, \phi(x_i) \rangle = \sum_{j=1}^n \alpha_j \langle \phi(x_j), \phi(x_i) \rangle = \sum_{j=1}^n \alpha_j K(x_j, x_i) = (K_S \alpha)_i.$$

Also

$$\|w^{\star}\|^2 = \left\langle \sum_{i=1}^n \alpha_i \phi(x_i), \sum_{j=1}^n \alpha_j \phi(x_j) \right\rangle = \sum_{i,j=1}^n \alpha_i \alpha_j K(x_i, x_j) = \alpha^{\top} K_S \alpha.$$

Thus the infinite-dimensional optimization problem is equivalent to

$$\min_{\alpha \in \mathbb{R}^n} \left[\frac{1}{n} \sum_{i=1}^n \text{loss}((K_S \alpha)_i; y_i) + \lambda \alpha^{\top} K_S \alpha \right].$$

All dependence on the sample enters only through the Gram matrix K_S .

2 Part B: ℓ_1 Rademacher Complexity

2.1 Problem 1. Convex Hulls

For any fixed realization of the signs $\sigma = (\sigma_1, \dots, \sigma_n)$, define

$$c_f(\sigma) := \frac{1}{n} \sum_{i=1}^n \sigma_i f(x_i).$$

Then

$$\sup_{h \in \text{conv}(\mathcal{F})} \frac{1}{n} \sum_{i=1}^n \sigma_i h(x_i) = \sup_{\substack{a_f \geq 0 \\ \sum_f a_f = 1}} \sum_{f \in \mathcal{F}} a_f c_f(\sigma).$$

The right-hand side is the maximum of a linear function over the simplex, so it is attained at an extreme point:

$$\sup_{\substack{a_f \geq 0 \\ \sum_f a_f = 1}} \sum_{f \in \mathcal{F}} a_f c_f(\sigma) = \max_{f \in \mathcal{F}} c_f(\sigma) = \sup_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n \sigma_i f(x_i).$$

Taking expectation over σ yields

$$\widehat{\mathfrak{R}}_S(\text{conv}(\mathcal{F})) = \widehat{\mathfrak{R}}_S(\mathcal{F}).$$

2.2 Problem 2. ℓ_1 Linear Predictors

Define the finite class

$$\mathcal{F}_0 = \{0\} \cup \{x \mapsto B\phi_j(x) : j \in [d]\} \cup \{x \mapsto -B\phi_j(x) : j \in [d]\}.$$

Every function in $\mathcal{F}_0$ takes values in $[-BR, BR]$ because $\|\phi(x)\|_\infty \leq R$.

Now let $h_w(x) = \langle w, \phi(x) \rangle$ with $\|w\|_1 \leq B$. Write

$$h_w(x) = \sum_{j=1}^d w_j \phi_j(x) = \sum_{j=1}^d \frac{|w_j|}{B} \text{sign}(w_j) B\phi_j(x).$$

If $\|w\|_1 < B$, add the missing weight

$$1 - \frac{\|w\|_1}{B}$$

to the zero function. Hence $h_w \in \text{conv}(\mathcal{F}_0)$. Therefore

$$\mathcal{H}_B^1 \subseteq \text{conv}(\mathcal{F}_0).$$

Using Problem 1,

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_B^1) \leq \widehat{\mathfrak{R}}_S(\text{conv}(\mathcal{F}_0)) = \widehat{\mathfrak{R}}_S(\mathcal{F}_0).$$

Massart's finite-class lemma now gives

$$\widehat{\mathfrak{R}}_S(\mathcal{F}_0) \leq BR \sqrt{\frac{2 \log |\mathcal{F}_0|}{n}} \leq BR \sqrt{\frac{2 \log(2d+1)}{n}}.$$

So

$$\widehat{\mathfrak{R}}_S(\mathcal{H}_B^1) \leq BR \sqrt{\frac{2 \log(2d+1)}{n}} = O\left(BR \sqrt{\frac{\log d}{n}}\right).$$

This is the desired logarithmic-in-dimension bound.

2.3 Problem 3. Comparison with Sparsity Bounds

Let $\ell(\hat{y}, y)$ be G -Lipschitz in $\hat{y}$ and bounded in $[0, 1]$. By contraction, the Rademacher complexity of the loss class is at most

$$G \hat{\mathfrak{R}}_S(\mathcal{H}_B^1).$$

Using the Week 7 uniform-convergence theorem together with Problem 2, there is a universal constant C such that, with probability at least $1 - \delta$ over $S \sim \mathcal{D}^n$, simultaneously for all $\|w\|_1 \leq B$,

$$L_{\mathcal{D}}^\ell(w) - L_S^\ell(w) \leq C \left(GBR \sqrt{\frac{\log(2d+1)}{n}} + \sqrt{\frac{\log(1/\delta)}{n}} \right).$$

2.3.1 Comparison with the Sparse VC Bound

The Week 4 sparsity bound says that for a k -sparse linear classifier,

$$L_{\mathcal{D}}^{0/1}(h) - L_S^{0/1}(h) \leq C' \sqrt{\frac{k \log(ed/k) + \log(1/\delta)}{n}}.$$

The two guarantees measure different notions of complexity.

- In the sparsity bound, the complexity is the support size k .
- In the ℓ_1 Rademacher bound, the complexity is the norm budget B together with the feature scale R , and the dimension enters only through $\log d$.

So the VC bound is *support-sensitive*, while the Rademacher bound is *scale-sensitive*. They are not interchangeable because they control different loss functionals.

The ℓ_1 bound does not directly control 0/1 error. To say anything about classification error, one needs an extra step: use a Lipschitz surrogate that upper-bounds or dominates 0/1 loss, such as hinge loss or the ramp loss from Week 7, and then convert small surrogate risk into a classification statement. Finally, the ℓ_1 bound can be informative when sparsity is not. Consider the dense vector

$$w = \left(\frac{B}{d}, \dots, \frac{B}{d} \right),$$

so $\|w\|_1 = B$ but its support size is

$$k = d.$$

If $d \gg n$, then the sparsity bound becomes

$$\Omega \left(\sqrt{\frac{d}{n}} \right),$$

which is vacuous. By contrast, the ℓ_1 bound is only

$$O \left(BR \sqrt{\frac{\log d}{n}} \right),$$

which can still be meaningful for moderate B .

3 Part C: Normalized ℓ_1 Margins and FixedBoost

3.1 Problem 1. Margin Generalization

Fix $w \neq 0$ with empirical normalized margin

$$\text{margin}_S(w) = \min_i \frac{y_i f_w(x_i)}{\|w\|_1} \geq \gamma.$$

Set

$$u := \frac{w}{\|w\|_1},$$

so $\|u\|_1 = 1$ and

$$y_i f_u(x_i) \geq \gamma \quad \text{for every } i.$$

Define the rescaled ramp loss

$$\psi_\gamma(z) := \min\{1, (1 - z/\gamma)_+\}.$$

It satisfies:

- $\mathbf{1}[z \leq 0] \leq \psi_\gamma(z)$ for all z ,
- ψ_γ is $(1/\gamma)$ -Lipschitz,
- $\psi_\gamma(z) = 0$ whenever $z \geq \gamma$.

Therefore

$$\frac{1}{n} \sum_{i=1}^n \psi_\gamma(y_i f_u(x_i)) = 0.$$

Now view u as an ℓ_1 -bounded linear predictor over the feature map

$$\Phi(x) := (g(x))_{g \in \mathcal{G}} \in \{-1, +1\}^{|\mathcal{G}|},$$

so

$$\|\Phi(x)\|_\infty = 1.$$

Apply the Part B.3 generalization-gap bound with

$$B = 1, \quad R = 1, \quad G = 1/\gamma,$$

and dimension $|\mathcal{G}|$. With probability at least $1 - \delta$, uniformly over all $\|u\|_1 \leq 1$,

$$L_{\mathcal{D}}^{\psi_\gamma}(u) - L_S^{\psi_\gamma}(u) \leq C \left(\frac{1}{\gamma} \sqrt{\frac{\log(2|\mathcal{G}| + 1)}{n}} + \sqrt{\frac{\log(1/\delta)}{n}} \right)$$

for a universal constant C .

Since the empirical surrogate loss is zero,

$$L_{\mathcal{D}}^{\psi_\gamma}(u) \leq C \left(\frac{1}{\gamma} \sqrt{\frac{\log(2|\mathcal{G}| + 1)}{n}} + \sqrt{\frac{\log(1/\delta)}{n}} \right).$$

Because ψ_γ upper-bounds 0/1 loss and $\text{sign}(f_u) = \text{sign}(f_w)$,

$$L_{\mathcal{D}}^{0/1}(\text{sign } f_w) \leq L_{\mathcal{D}}^{\psi_\gamma}(u).$$

Hence, with probability at least $1 - \delta$,

$$L_{\mathcal{D}}^{0/1}(\text{sign } f_w) \leq C \left(\frac{1}{\gamma} \sqrt{\frac{\log(2|\mathcal{G}| + 1)}{n}} + \sqrt{\frac{\log(1/\delta)}{n}} \right).$$

Up to constants, this is the desired margin bound in terms of $|\mathcal{G}|$, n , γ , and δ .

3.2 Problem 2. FixedBoost Implication

The black-box theorem says that under the weak-learning condition

$$\forall D \in \Delta_n, \exists g \in \mathcal{G} \quad \text{with weighted error} \leq \frac{1}{2} - \frac{\gamma}{2},$$

FixedBoost returns w_T after

$$T = O\left(\frac{\log n}{\gamma^4}\right)$$

rounds such that

$$\min_i y_i f_{w_T}(x_i) \geq \frac{\gamma}{2} \|w_T\|_1.$$

Equivalently,

$$\text{margin}_S(w_T) \geq \frac{\gamma}{2}.$$

Applying Problem 1 with margin parameter $\gamma/2$, we obtain that with probability at least $1 - \delta$,

$$L_D^{0/1}(\text{sign } f_{w_T}) \leq C \left(\frac{1}{\gamma} \sqrt{\frac{\log(2|\mathcal{G}| + 1)}{n}} + \sqrt{\frac{\log(1/\delta)}{n}} \right)$$

for another universal constant C .

So the classifier returned by FixedBoost generalizes at a rate controlled by the achieved normalized ℓ_1 margin, not by the number of rounds directly. The round count only enters indirectly through the theorem that guarantees a margin of order γ .

3.3 Problem 3. Weak Learning and Normalized ℓ_1 Margin

Let $m := |\mathcal{G}|$ and write the sign matrix as

$$A \in \{-1, +1\}^{n \times m}, \quad A_{ig} = y_i g(x_i).$$

3.3.1 Weak-Learning Edge

For a distribution $D \in \Delta_n$ over the sample, the weighted error of g is

$$\sum_{i=1}^n D_i \mathbf{1}[g(x_i) \neq y_i] = \frac{1}{2} - \frac{1}{2} \sum_{i=1}^n D_i y_i g(x_i) = \frac{1}{2} - \frac{1}{2} (D^\top A)_g.$$

Therefore the edge of g under D is

$$\frac{1}{2} - \text{err}_D(g) = \frac{1}{2} (D^\top A)_g.$$

So the best weak-learning edge of the sample is

$$\gamma_{\text{weak}} = \frac{1}{2} \min_{D \in \Delta_n} \max_{g \in \mathcal{G}} (D^\top A)_g.$$

3.3.2 Normalized Margin

Because $\mathcal{G}$ is symmetric, any negative coefficient on g can be absorbed as a positive coefficient on $-g$. So when maximizing normalized ℓ_1 margin, we may restrict without loss of generality to nonnegative coefficients whose sum is 1. Thus

$$\max_w \text{margin}_S(w) = \max_{u \in \Delta_m} \min_{i \in [n]} (Au)_i.$$

Now

$$\min_{i \in [n]} (Au)_i = \min_{D \in \Delta_n} D^\top Au,$$

because minimizing a linear function over the simplex picks out the smallest coordinate. Hence

$$\max_w \text{margin}_S(w) = \max_{u \in \Delta_m} \min_{D \in \Delta_n} D^\top A u.$$

By von Neumann's minimax theorem,

$$\max_{u \in \Delta_m} \min_{D \in \Delta_n} D^\top A u = \min_{D \in \Delta_n} \max_{u \in \Delta_m} D^\top A u.$$

For fixed D , maximizing over the simplex Δ_m picks the largest coordinate:

$$\max_{u \in \Delta_m} D^\top A u = \max_{g \in \mathcal{G}} (D^\top A)_g.$$

Therefore

$$\max_w \text{margin}_S(w) = \min_{D \in \Delta_n} \max_{g \in \mathcal{G}} (D^\top A)_g.$$

Comparing with the expression for γ_{weak} , we obtain the exact relationship

$$\max_w \text{margin}_S(w) = 2\gamma_{\text{weak}}.$$

3.4 Problem 4. Connection to Homework 6

Homework 6 gave a one-sided implication: a particular sparse comparator w^* with large normalized ℓ_1 margin forces every reweighting to admit a coordinate rule with nontrivial edge. The comparator there was a *specific sparse witness*, and the final generalization step turned T rounds of boosting into a T -sparse linear predictor, then used a sparse-VC argument on the activated support. The present margin viewpoint refines both parts of that story.

- The comparator changes from one chosen sparse witness w^* to the *best possible normalized ℓ_1 margin* over all ensembles of base rules.
- The weak-learning statement becomes exact: by Problem 3, the best weak edge is exactly half of the best normalized ℓ_1 margin.
- The generalization argument changes from a support-size VC bound to a norm-based Rademacher bound. The resulting complexity depends on

$$\frac{1}{\gamma} \sqrt{\frac{\log |\mathcal{G}|}{n}},$$

not on the number of nonzero coordinates or the number of rounds directly.

So Week 7 replaces the coarse “how many rules were used?” viewpoint by the sharper “what normalized ℓ_1 margin was achieved?” viewpoint.

4 Part D: FixedBoost Margin Experiment

4.1 Problem 1. Implementation

The implementation is in the relative path `code/fixedboost_experiment.py`, and the numerical output is in `code/fixedboost_summary.json`. I implemented FixedBoost directly on the sign matrix

$$A_{ig} = y_i g(x_i) \in \{-1, +1\}.$$

The loop differs from Homework 6's AdaBoost harness in three ways:

- the step size is fixed at $\eta = 0.1$ instead of using the adaptive AdaBoost coefficient,
- the algorithm works directly with the matrix A rather than recomputing correlations from raw features and labels,
- the tracked quantity of interest is the normalized empirical ℓ_1 margin together with the LP optimum, rather than only training error or exponential loss.

Because the base class consists of signed coordinate predictors, the optimal normalized ℓ_1 margin can be written as the linear program

$$\max_{\rho, u} \rho$$

subject to

$$Au \geq \rho \mathbf{1}, \quad u \geq 0, \quad \sum_g u_g = 1.$$

The simplex constraint is enough because the signed base class is symmetric, so negative coefficients can be folded into positive weights on negated predictors.

4.2 Problem 2. Optimal-Margin Comparison

I used two deterministic instances.

4.2.1 Large-Margin Instance

The easy instance is the majority-of-three sample:

$$x \in \{-1, +1\}^3$$

ranges over all 8 Boolean patterns, and the label is

$$y = \text{sign}(x_1 + x_2 + x_3).$$

The LP returns the exact optimal normalized margin

$$\rho_{\text{easy}}^* = \frac{1}{3}.$$

FixedBoost reaches half of this value by round 3, reaches 90% of it by round 3, and in fact hits the exact optimum at round 3. By round 120 the run ends exactly at

$$\text{margin}_S(w_{120}) = \frac{1}{3}.$$

This large margin comes from the structure of A : the three informative coordinate columns can be averaged so that every row keeps signed score at least 1 out of total ℓ_1 mass 3.

4.2.2 Small-Margin Instance

The hard instance reuses the Homework 6 sign-matrix construction with $m = 3$, so the underlying Boolean dimension is

$$d = 2m + 1 = 7.$$

I take the 7 sign vectors from that construction, label every example by $+1$, and repeat the rows with multiplicities

$$q = (1, 1, 1, 2, 2, 4, 4).$$

This yields a 15-example sample whose signed-coordinate matrix is hard for weak learning. For this instance the LP returns

$$\rho_{\text{hard}}^* \approx 0.0303.$$

FixedBoost reaches half of this optimum by round 114, reaches 90% of it by round 510, and by round 1500 attains

$$\text{margin}_S(w_{1500}) \approx 0.0293,$$

leaving a gap of only about

$$9.7 \times 10^{-4}.$$

This small optimum reflects the structure of A : there are reweightings under which every signed coordinate looks only barely better than random, so no convex combination can make every row strongly positive at once.

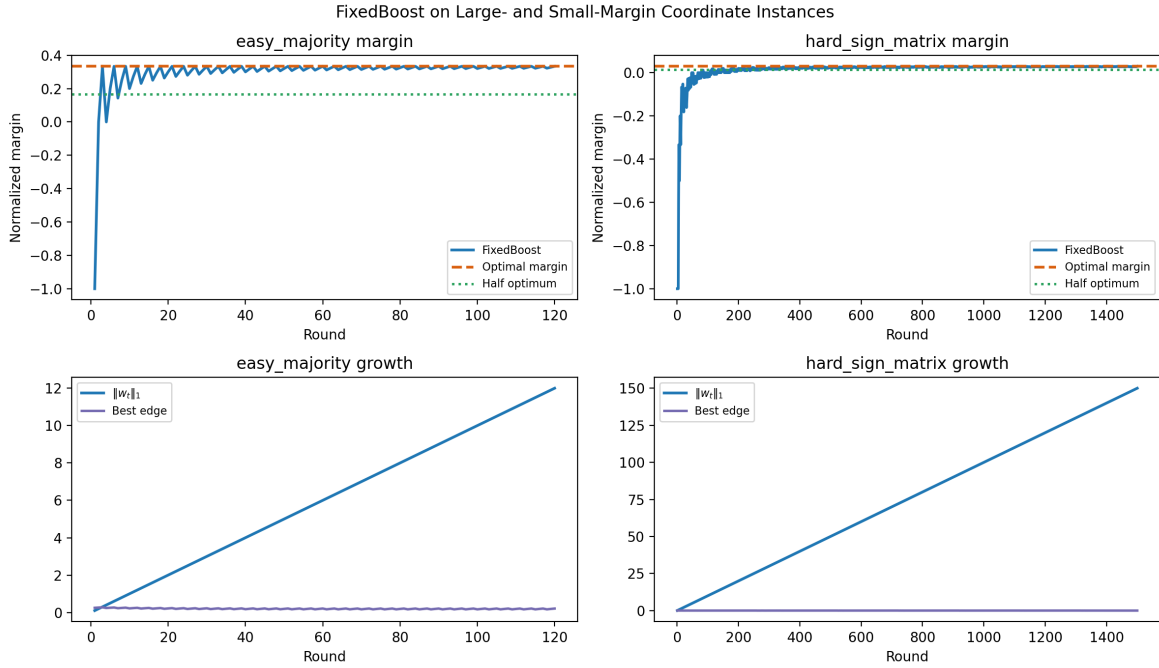


Figure 1: FixedBoost normalized-margin trajectories and ℓ_1 growth on the easy and hard instances.

4.3 Problem 3. Interpretation

Yes: on both instances, FixedBoost drives the normalized margin toward the LP optimum.

On the easy majority-of-three instance, convergence is essentially immediate: the optimum is exactly $1/3$, and the algorithm reaches it after three coordinate updates. The behavior is almost periodic after that because the same three informative columns keep reappearing.

On the hard sign-matrix instance, convergence is much slower but still clear. The optimum is only about 0.0303 , and the run gets to about 0.0293 by round 1500. The trajectory is therefore approaching the optimal value, but only after a long transient.

By Problem C.3, the theorem parameter γ is exactly the optimal normalized margin ρ^* . So the comparison with Part C.2 is especially revealing. The theorem-scale proxy

$$\frac{\log n}{(\rho^*)^4}$$

is about

$$168.4$$

for the easy instance and about

$$3.21 \times 10^6$$

for the hard instance. Empirically, however, the easy run reaches the optimum in 3 rounds and the hard run reaches 90% of optimum in 510 rounds. So the theorem is qualitatively right but quantitatively very loose.

There are at least two reasons for the looseness.

- The theorem only guarantees a margin of order $\gamma/2$ after $O(\log n/\gamma^4)$ rounds; it does not try to predict when the iterates get close to the true optimum.
- Problem C.3 is a minimax statement over *all* reweightings D . The distributions actually visited by FixedBoost can be much friendlier than the adversarial worst-case distribution that defines ρ^* .

So the experiment shows that the margin theorem is conservative: it gives a valid worst-case certificate, but it can be far slower and less precise than the dynamics observed on concrete samples.

5 Part E: Agent Skill Reflection

5.1 Problem 1. Existing Workflow

I reused the existing repo workflow from `AGENTS.md`: check that the `ai4lt` Conda environment is active, use `pdflatex` for builds, keep one companion statement file per homework, store code in `hw7/code`, store plots in `hw7/figures`, and keep long code out of the PDF by referencing the relative path instead.

5.2 Problem 2. Skill Development

I did not modify the local skill registry for this assignment, so the concrete skill I would add is a `lt-margin-experiment` skill. It would automate four recurring tasks from this homework:

- build the sign matrix A from labeled Boolean data,
- solve the normalized-margin LP,
- run a fixed-step boosting loop while logging margin and ℓ_1 growth,
- regenerate plots and a summary JSON so the LaTeX writeup can quote verified numbers.

5.3 Problem 3. Verification

The main coding-agent failure modes here are: mixing up the margin normalization by $\|w\|_1$, forgetting that symmetry lets negative coefficients be folded into positive weights on $-g$, solving the wrong LP, or reporting theorem quantities that do not match the implemented experiment.

I checked the assignment in several ways. For the math, I verified the minimax identities and the kernel/Rademacher derivations line by line against the exact problem statements and the Week 7 slides. For the code, I checked that the LP optimum on the easy instance is exactly $1/3$, which is analytically expected, and I confirmed that the hard instance produces a much smaller optimum. For the plots and reported claims, I regenerated `code/fixedboost_summary.json` and the PNG figure from the script, then copied only those saved values into the final writeup. Finally, I rebuilt the PDF with `pdflatex` to make sure the report and the repository artifacts are synchronized.